



UNIVERSITY
OF TRENTO - Italy

Dipartimento di Ingegneria e Scienza dell'Informazione



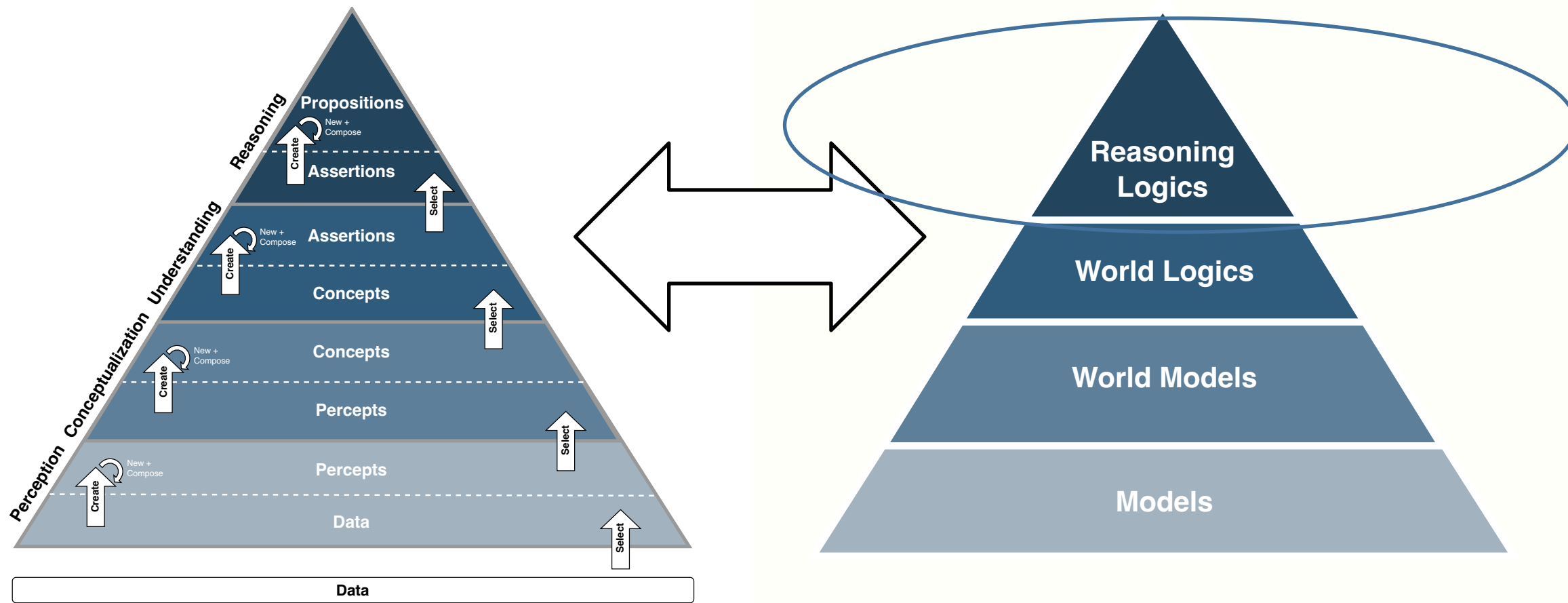
Language Logics

(HP2T)

Language Logics

- **Introduction**
- What LoDE does / does not represent
- Formulas, assertions, propositions
- Language Entailment
- Language Logics
- Truth, Falsity and IDK in world logics
- Logical Entailment
- Properties of language Logics
- Language Logics diversity
- Key notions

The knowledge Pyramid and the Logics pyramid (recall)



World entailment (Recall)

Definition (World entailment) Let $W = \langle L_a, D, I_a \rangle$ be a world model. Let L_a be an assertional language. Then $\models_{L_a}$ is a **world entailment relation**

$$\models_{L_a} \subseteq D \times L_a$$

such that

$$M \models_{L_a} T_a$$

$M = \{f\} \subseteq D$ is a **model of** $T_a \subseteq L_a$. We say that M **entails** T_a . We write $M \models T_a$ instead of $M \models_{L_a} T_a$ when no confusion arises.

World Logics (Recall)

Proposition (World Logic). Let $W' = \langle L_{a'}, D', I_{a'} \rangle$ be a World Model. Then a **World Logic** L_W for W is defined as

$$L_W = \langle W, \models_W \rangle$$

with $W = \langle L_a, D, I_a \rangle$ and $L_{a'} \subseteq L_a$, $D' \subseteq D$ and $I_{a'} \subseteq I_a$.

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LoDE – representation of Disjointness

Observation (Representation of disjointness). LoDE does not allow to represent (as part of the language), as an assertion, that **two LoE assertions are disjoint** (that is, different), this being also the case in the corresponding world models. This also applies to LoD /LoC assertions KG theories and world models.

Example (Representation of disjointness). Consider the assertions “my T-shirt is green” and “my T-shirt is grey”. Their relation cannot be formalized in LoDE, for instance as

$$\text{Color}(\text{myTshirt}\#1, \text{green}) \perp \text{Color}(\text{myTshirt}\#1, \text{grey})$$

... and computed as a consequence, via expansion, of the general LoDE description

$$\text{Color}(\text{object}, \text{green}) \perp \text{Color}(\text{object}, \text{grey})$$

applied to the LoE assertion

$$\text{object}(\text{myTshirt}\#1)$$

Observation (Representation of disjointness). LoDE does not allow either to use the negation operator “ $\neg$ ” in LoE (and LoD/ LoC) assertions. It is impossible to assert negative facts at the entity, language and knowledge level.

Observation (Representation of disjointness). LoDE does not allow assertions about what is not explicitly represented in models / analogic representations of the world.

Intuition (Representation of disjointness). What is the intuitive meaning of the pseudo-assertion above?

LoDE – representation of partial knowledge

Observation (Representation of partial knowledge). LoDE allows to state what is the case. But , in particular in the case of LoE entity level assertions, it does not allow to represent, as an assertion, anything about “the rest”, namely, what is **not represented explicitly** and is therefore **unknown**.

Example (Representation of partial knowledge). Consider the assertion “my T-shirt is green”. Consider now the following two assertions:

- “my t-shirt is grey”?
- “my t-shirt is grey or green”?

They cannot be part of the same world description. There is a key distinction between **partial knowledge** and **negative knowledge**, which is not captured by LoDE EGs.

Observation (Representation of partial knowledge). LoDE does not allow to use the disjunction operator “ $\sqcup$ ” in LoE (and LoD/ LoC) assertions. As for disjointness, it is impossible to assert partial facts at the entity, language and also knowledge level.

Observation (Closed World Assumption - CWA). CWA does not allow to reason about partiality. It assumes complete knowledge reducing “not knowing” to “knowing not”.

Intuition (Representation of partial knowledge). What is the intuitive meaning of the pseudo-assertion above?

LoDE – representation of Logical Consequence

Observation (LoDE does not allow for Logical Consequence). LoDE allows to state what is the case. But it does not allow to represent, as a new assertion, **which assertions follow from** (that is, are consequences), **assertions which are known to be True /False.**

Example (Logical Consequence). If one knows the following facts

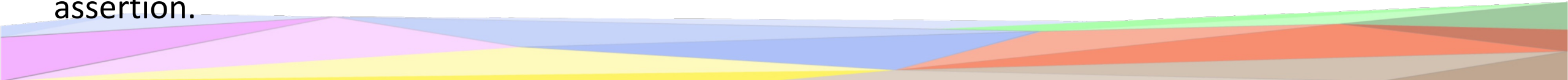
- Whenever Fausto is not at home or he sleeps his children make a lot of noise
- Fausto's children are quiet

then one can conclude that

- Fausto is at home or he sleeps

Observation (“Whenever Fausto is not at home or he sleeps his children make a lot of noise”). This sentence cannot be expressed in LoDE. In fact it does not follow the BNF for assertions. For two reasons: (1) entities are mentioned (2) these “pseudo-”assertions are nested

Intuition (Logical Consequence). Logical consequence is logical entailment applied to assertions. The fact that an assertion is a logical consequence of another assertion cannot be represented as a LoDe assertion.



LoDE formation rules – BNF (RECALL)

$$\begin{array}{l} \langle a_{\text{LoDE}} \rangle ::= \langle a_{\text{LoE}} \rangle \\ \qquad \qquad \qquad \langle a_{\text{LoD}} \rangle \\ \qquad \qquad \qquad \langle a_{\text{LoC}} \rangle \\ \langle a_{\text{LoD}} \rangle ::= \langle a_{\text{LoDef}} \rangle \\ \qquad \qquad \qquad \langle a_{\text{LoDes}} \rangle \end{array} \quad \left| \right.$$

***Assertions do not take
assertions
as arguments!!!!***

No reasoning on assertions

Observation (LoDE formation rules). The set of the LoDE Formation rules are the union of sets of formation rules of LoE, LoD and LoC.

What LoDE does / does not represent

LoDE ...

- Provides us with the means to represent all the world models, i.e., EGs, ETGs, CGs and the world models composed from them;
- Provides us (LoC) with the means to represent and reason about the commonsense meaning of language (concepts, concept definitions);
- Provides us (LoD) with the means to represent and reason about commonsense knowledge (etypes, etype definitions and descriptions);
- Provides us (LoE) with the means to reason about entities based on the knowledge encoded in LoC and LoD assertions (LoE assertions);
- Provides us with the means to compose LoC, LoD and LoE reasoning;
- Does not allow us to represent and to reason about what the consequences of the assertions which are represented in world models (similarity and dissimilarity) and entities in particular.

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Logics – Intuition

Observation (World logics). World logics formalize how **truth in a model** can be reasoned about in a (logical) theory (that is, a linguistic representation of the world). They are the key basic element, via the entailment relation, for the formalization of **(logical) reasoning**.

Observation (Language logics). Logical reasoning is linguistic reasoning, that is, reasoning in a predefined language. Logical reasoning is implemented using **(language) logics** which allow **to draw conclusions** from the **true facts computed by world logics**.

Observation (World and language logics). Language logics use world logics as **oracles** which provide information about what is true/ false in the intended model. Language logics implement reasoning about truth (i.e, the truth asserted by assertions).

Assertions

Definition (Assertion). An **Assertion** a is an **atomic sentence**, that is, a sentence which cannot be decomposed into simpler sentences, which unambiguously describes a single **fact**. For any assertion a we have

$$a \in L_a = \{a\}$$

Observation (From facts to assertions) The mapping from facts to assertions is in full control of the modeler. We have the following:

- For formation rules there is no specific recipe. The general idea is that they should generate a syntax which makes it as easy as possible for people to understand the underlying facts (see above);
- A special case, see above, is when there is no need of formation rules.

Types of assertions (reprise)

Definition (Assertion). A **fact**, as denoted by an **assertion** a has one of the following five forms

- Assertion starting fact: $u_i \in C_j$,
- Assertion starting fact: *Tuple of Units memberOf relation*: $\langle u_1, \dots, u_n \rangle \in R^n$,
- Assertion starting fact: *Class subsetOf Class*: $C_i \subseteq C_j$,
- Assertion starting fact: *Relation subsetOf relation*: $R_i^n \subseteq R_j^n$
- Assertion starting fact: *Relation subsetOf tuple of classes and viceversa*:
 - $R^n \subseteq C_1 \times \dots \times C_n$
 - $C_1 \times \dots \times C_n \subseteq R^n$

Different assertional languages differ in the choice of facts to be asserted. See examples below.

Judgements

Intuition (Judgement by Merriam-Webster on line). A **judgement** is the process of forming an opinion or evaluation by discerning and comparing.

Observation (Judgement). What are judgements, judgements of? ... **of assertions!**

Intuition (Economics Dictionary of Arguments on line). A **judgment** differs from a statement (an “assertion” in our earlier terms) in that it also asserts the truth of its content.

Examples (Judgement):

- It is true that the door is open (but also: It is true/false that it is true the door is open)
- It is false that the door is open (but also: It is true/false that it is false the door is open)

... as different from the assertions

- The door is open
- The door is closed

Observation (Assertion and judgement). An assertion describes (falsely or truly) what is the case. A judgement is a (meta)statement about the truth of an (object) assertion. We have

- A judgement of falsity / truth about a false assertion
- A judgement of truth / falsity about a true assertion

Propositions

Notion (Aristotle). A **proposition** is a sentence which affirms or denies a predicate of a subject.

Intuition (Proposition). A **proposition is a judgment about an assertion**, and more in general, **a judgement about a proposition**. We have two types of propositions:

- A certain assertion (proposition) is False
- A certain assertion (proposition) is True

Intuition (Value of propositions). A proposition can denote true or false. This is the intended model. A proposition

- can be either true or false,
- it must be one or the other,
- and it cannot be both.

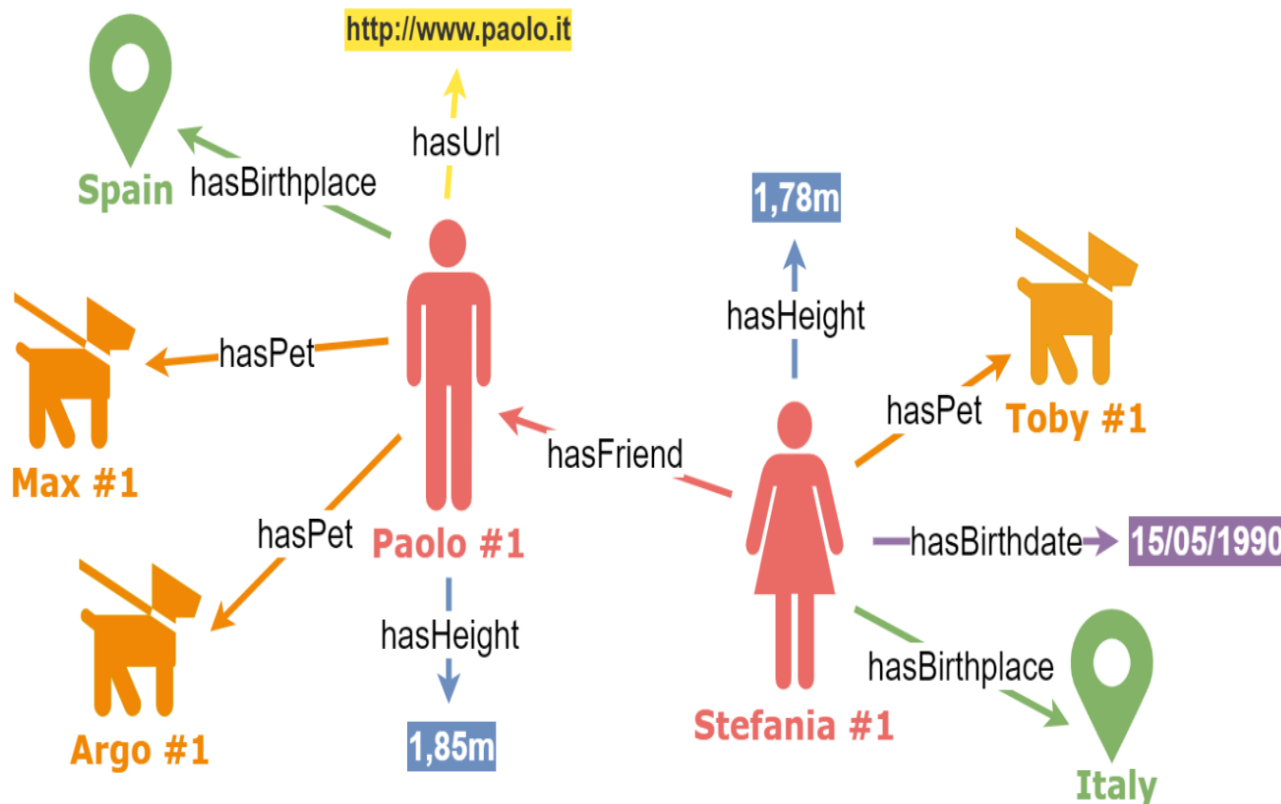
Observation (Proposition, IDK). We do not allow propositions to be IDK. We need to deal with this case when this is the case in the reference LoDE KG theory.

Types of propositions

Intuition (Atomic and complex propositions). There are two types of propositions:

- **Atomic propositions**, which are not further decomposable and which are either true or false, and
- **Complex propositions**, composed of atomic propositions, whose truth value depends on the truth value of the atomic propositions of which they are composed.

Atomic propositions – A LoDE EG example



Which of the following propositions are intuitively true?

- $HasFriend(Stefania\#1, Paolo\#1) = T$
- $HasFriend(Stefania\#1, Paolo\#1) = F$
- $Hasheight(Stefania\#1, 2m) = T$
- $Hasheight(Stefania\#1, 2m) = F$

Person

String

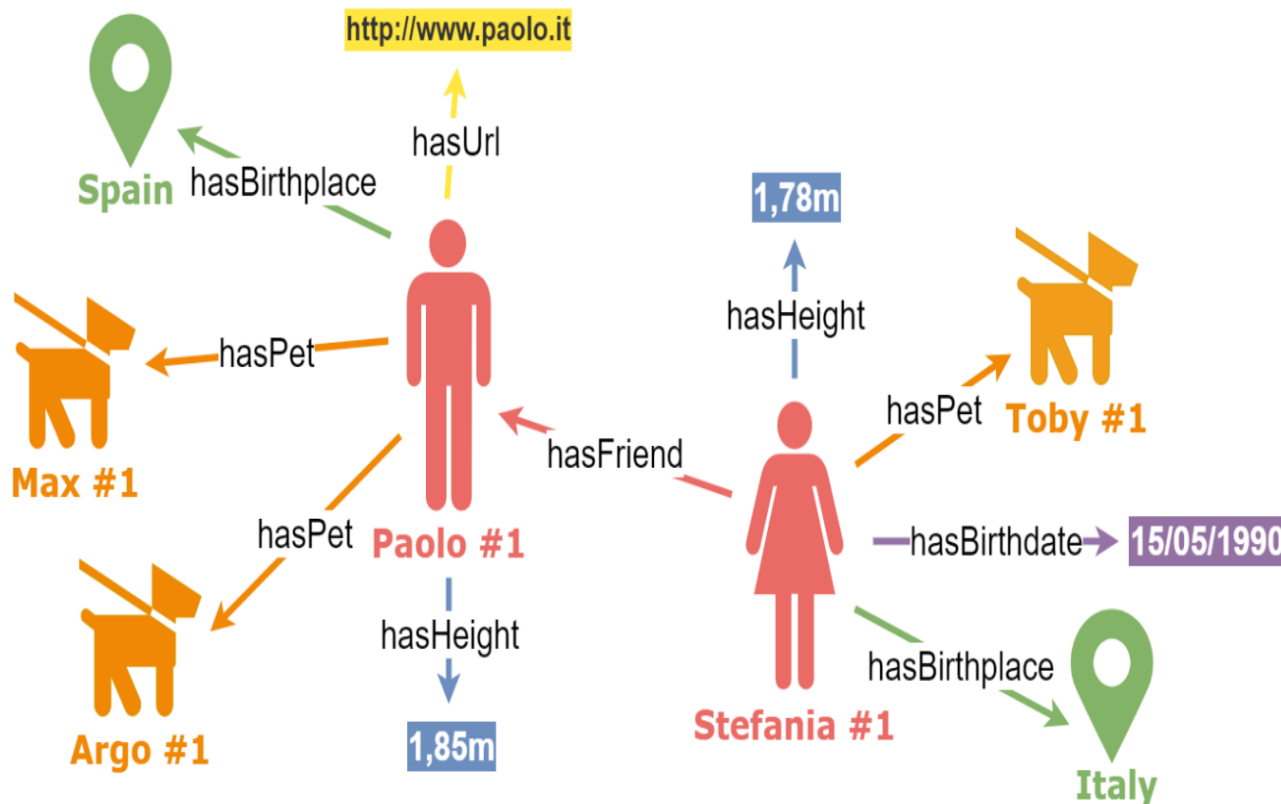
Dog

Place

Real

Date

Complex propositions – A LoDE EG example



Which of the following complex propositions are intuitively true?

- $HasFriend(Stefania\#1, Paolo\#1) = F$
and $HasHeight(Stefania\#1, 2m) = F$
- $HasFriend(Stefania\#1, Paolo\#1) = F$
or $HasHeight(Stefania\#1, 2m) = T$

No need of the LoDE EG when computing the truth value of complex propositions! We could have written the two propositions above as:

- C and B
- A or B

with $C=F$, $A= F$, $B= T$.

Complex propositions – A language-only example

Example (Complex propositions). Propositions composed out of assertions are in "quotes". Within quotes, the judgement is underlined. Natural language connectives are underlined. and, or, not, if ... then, iff, and not have their intuitive meaning.

A = "It is true that Stefania#1 is a friend of Paolo#1" ;

B = "It is false that Stefania#1 is a friend of Paolo#1" ;

C = "It is true that today it rains";

D = "It is false that Paolo#1 is a woman";

E = "It is true that Paris is the capital of Italy"

G = "It is false that it is true that Paris is the capital of Italy"

1. A
2. B
3. A and not C
4. A or D
5. (A or D) and E
6. If not E then G
7. A and not A
8. A or not A
9. E iff not G

Exercise. Compute the truth value of the propositions on the right with respect to the LoDE EG.

Terminology – formulas

Terminology (Formula, well-formed formula, wff). All logics rely on a language, defined in terms of a type 2 Chomsky grammar, composed of an alphabet and a set of formation rules, an interpretation function and a domain of interpretation.

Languages can be distinguished in terms of the objects which constitute their domain of interpretation. We call **formula**, or **well-formed formula**, or **wff** any element of given language which is correctly formed, starting from the alphabet and using the formation rules, independently of their domain of interpretation. Notationally, we write

$L = \{w\}$ to mean that the language L is a set of wffs w ,

Terminology (Assertion). Assertions, distinct in atomic and complex assertions, are formulas which describe facts of composition of facts as they occur in the intended model. Notationally, we write

$L = \{a\}$ to mean that the language L is a set of assertions a

Terminology (Proposition). Propositions, distinct in atomic and complex propositions, are formulas which describe what is true in the intended model. Notationally, we write

$L = \{p\}$ to mean that the language L is a set of propositions p

Observation (formula). The distinction among the different types of formulas (e.g., assertion, vs. proposition) is based on what they denote, and it is independent of the specific alphabet and formation rules.

Logics represent positive and negative knowledge

Observation (Representing LoDE positive and negative knowledge). The (explicit) positive and the (implicit) negative knowledge of a LoDE representation is encoded by:

- representing positive knowledge (a true assertion) with the proposition which encodes it (that is: “it is true that ...”)
- representing negative knowledge (a false assertion) with a proposition which encodes it (that is: “it is false that ...”)
- requesting that any atomic proposition “P” to take the value True or False, but not both;
- requesting that the domain D allows, for any assertion, to assert that this assertion is either True or False.

Observation (Contents of a domain). A domain D needs only to contain two percepts, that is, the pair <True, False>

Observation (Symbol for negation). The above approach requires the existence of $2 \cdot N$ atomic propositions, where N is the number of assertions. The number of atomic propositions can be reduced to N but allowing the symbol not and the complex proposition where “not P”, where “P” is an atomic proposition.

Logics represent partial knowledge

Observation (Representing and reasoning about partial knowledge). Language logics model partiality (1) by requesting that models are complete, that is, that they assign a truth value to each atomic proposition and (2) by allowing a formula has multiple models

Observation (Theory partiality). Theories are allowed to be partial (as in world logics).

Example (LoP complex propositions). Consider the previous example.

A = "It is true that Stefania#1 is a friend of Paolo#1" ;

B = "It is false that Stefania#1 is a friend of Paolo#1" ;

C = "It is true that today it rains";

1. A

2. B

3. A or not C

The theory containing the three propositions on the right will have 2,2,6 out of 8 possible models. All of them contain all truths defined by the input theory.

Observation (Contents of a model). A model M is completely described by as many elements, that is truth values, as there are atomic propositions in the language.

Logics represent partial knowledge (continued)

Observation (Reference model partiality). As discussed earlier on, reference models are most often incomplete and, as such, they are usually defined by a theory which is incomplete. This is allowed by world logics but not in (language) logics. An incomplete reference model, as it was allowed in world logics corresponds here to a set of models which (see above on theory partiality):

- All share the true propositions defined by the (incomplete) reference model
- Cover all possible combinations of truth value assignments to all propositions whose truth is unknown. The number of models doubles for every proposition whose truth value is unknown

Example (On the incompleteness of reference models). Consider the previous example.

- | | |
|--|-----------------------------|
| A = " <u>It is true that</u> Stefania#1 is a friend of Paolo#1" ; | 1. A |
| B = " <u>It is false that</u> Stefania#1 is a friend of Paolo#1" ; | 2. B |
| C = " <u>It is true that today it rains</u> "; | 3. A <u>or</u> <u>not</u> C |

A theory containing the three propositions on the right will have 1 out of 8 possible models.

Observation (Maximal theory). In a language logic, a theory, to be maximal, must assign a truth value to all (atomic) propositions allowed by the language. This is different from world logics.

Logics represent Logical Consequence

Example (Logical Consequence).

If one knows the following two propositions

- IF (It is true that whenever Fausto is not at home) OR (It is true that he sleeps) THEN (It is true that his children make a lot of noise)
- It is true that Fausto's children are quiet

then one can conclude that

- (It is true that Fausto is at home) OR (It is true that he sleeps)

Intuition (Logical Consequence). Logical consequence is logical entailment applied to propositions, as represented in language logics

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Language Entailment

Notion 6 (Entailment) Let $W = \langle L_a, D, I_a \rangle$ be a world model. $L_W = \langle W, \models_{L_a} \rangle$ be a world logic for W . Let $L = \{w\}$ be a language, with $L_a \subseteq L$. Let $M = \{f\} \subseteq D$ be a set of facts. Let $T \subseteq L$ be a theory. Then $\models_L$ is an **entailment relation** that associates the facts in M with the elements in T , in formulas

$$\models_L \subseteq D \times L, \text{ also written } M \models_L T, (*)$$

subject to the constraint that for all assertions $a \in L_a$,

$$\text{if } M \models_{L_a} a \text{ then } M \models_L a \quad (**)$$

We also say that M **entails** T and write $M \models T$ when no confusion arises.

Observation ($L_a \subseteq L$). Usually the assertions $a \in L_a$ get rewritten, to assertions $a' \in L$, under the guarantee of a one-to-one mapping between the two notations.

Observation ($L_a \subseteq L$). $(**)$ holds only in one direction because of the partiality of world logic models. Remember that we have restricted ourselves to complete logic models.

Entailment – observations

Observation 1 (Entailment, reasoning). The definition of entailment is made based on a theory $T \subseteq L$, where there exists a suitable L_a , with $L_a \subseteq L$. The key intuition is that of extending a reference assertional language to allow for formulas which are not necessarily assertions and, then, to ask about the truth of these formulas.

Observation 2 (Entailment, reasoning). Entailment formalizes the intuitive notion of reasoning. It links what one asserts as being the case with what is true in the model. There are multiple notions of entailment, formalizing different notions of reasoning, even for the same world model, with wildly different properties, still maintaining the properties listed before.

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Logics and logic representations

Definition (Logic). Given a **world model** $W = \langle L_a, D, I_a \rangle$ and a **world logic** $L_W = \langle W, \models_{L_a} \rangle$, a **logic** L_L for L_W is defined as

$$L_L = \langle W, \models_L \rangle$$

where $L_a \subseteq L$, and $\models_L$ is an **entailment relation**.

Definition (Logic representation). Given a logic $L_L = \langle W, \models_L \rangle$, a (**logic representation** is defined as)

$$R = \langle T, M \rangle$$

with

$$\begin{aligned} M &\subseteq D \\ T &= \{w\} \subseteq L \end{aligned}$$

where M and T are, respectively, a **model** of T and a **theory** of M in L_L .

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Truth values of propositions

Intuition (Truth / falsity of atomic propositions). The truth / Falsity of atomic propositions is computed by “asking” a World Logic, the one describing the model described by the assertion whose truth is evaluated. The answer from the World Logic can be **True**, **False**, or **I don't know**.

Intuition (Truth / falsity of complex propositions). The truth / Falsity of complex propositions is computed from the truth value of atomic propositions, based on the entailment relation of the (language) logic. They depend on a proposition to be **True** or **False**.

Observation (truth of atomic vs complex formulas). The truth value of atomic propositions depends on what is the case in the underlying world model and logics (the language they use to describe the world model). The truth value of complex propositions does not depend on the world model. It only depends on the truth value of atomic propositions, independently of what the underlying assertions assert.

Observation (Reasoning). Reasoning is **linguistic** and independent of what is the case in the world. It only depends on how truth values compose in a statement. In this sense, it is **universal**, meaning by this, **independent of the world as we perceive it** (in our mental analogical representation) and also **independent of the world we describe it** (in our reference world model and logic).

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Logic – Correctness, completeness (same as world logics)

Definition (Logic Correctness and completeness). We have five levels of correctness / completeness

- Language correctness/completeness: relates language to domain of interpretation
- Interpretation correctness/completeness: relates assertions to facts
- Theory correctness/completeness: relates theory to model, or, said in another way, theorems to true facts
- Entailment correctness/completeness: relates model to (a new) theory, or, said in another way, axioms to theorems / true facts
- Decision procedure correctness/completeness: relates theorems to computable theorems

Exercise (Logic Correctness and completeness). Provide formal definitions of the five notions of correctness / completeness provided above.

What Logics do / do not represent

Logics ...

- Provide us with the means to represent linguistically and reason about the knowledge encoded in a LoDE representation;
- Provide us with the means to represent and reason about negative knowledge, as implicit in a LoDE representation;
- Provide us with the means to represent and reason about partiality, as implicit in a LoDE representation;
- Provide us with the means to draw consequences from what is known via logical consequence, as implicit in a LoDE representation;

Observation (World versus language logics). Differently from world logics, language logics do not have a graphical representation of the knowledge they encode (!). This is because their models are not analogical but, rather, linguistic.

Observation (World versus language logics). Language logics allow to represent and reason about what we do not directly perceive and that, therefore, cannot be represented in an analogical representation. They allow to represent and reason about what we learn it is the case by experience.

Logics– observations

Observation (Opinions/ reasoning fallacies / mistakes). The definition of language logics provides a partial explanation of why there are different opinions, reasoning fallacies or mistakes. Being reasoning completely abstracted from the world, any variation in the computation of the truth value of an atomic proposition, or even it in the reference world logics propagates to the entire reasoning path till a check of the truth value of the conclusion in the underlying world model is done (**reality check**).

Example (Reasoning fallacies). Some examples as from the first lecture

- Misconceptions
- Overgeneralization
- mental filtering
- Catastrophism
- Bias

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Logics

Observation (Logics). We will study the following two logics:

- **LoP.** The **Logic of Propositions**, also called **Propositional Logic**. It allows to reason about and to draw consequence from **propositions**, that is, from judgements about what is true and what is false;
- **Lol.** The **Logic of Interaction**, also known as the **First order Logic**. This logic allows for the use of variables, **existential** and **universal quantification**. It approximates the expressivity of the language used in natural language interactions (towards LLMs).

Observation (Lol). Because of our focus on Computer Science, we will focus on finite domains.

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Key notions

- World representation
- World model
- World logic
- (Language) Logic
- Assertions, judgments, propositions, formulas
- Atomic and complex propositions
- Knowledge about disjointness
- Negative knowledge
- Partial knowledge
- Logical consequence / entailment
- Logic entailment properties
- Reflexivity, cut, compactness, (non) monotonicity



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